Lesson Planning

Complete Lesson Plan Book

Class: Class 9

Subject: Chemistry

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Fundamentals of Chemistry

Unit 2: Structure of Atoms

Unit 3: Periodic Table and Periodicity

Unit 4: Structure of Molecules

Unit 5: Physical States of Matter

Unit 6: Solutions

Unit 7: Electrochemistry

Unit 8: Chemical Reactivity

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Unit 1: Fundamentals of Chemistry

Learning Objectives

• Define chemistry and name its main branches.
• Distinguish between element, compound and mixture.
• Define atomic number, mass number and isotopes.
• Calculate relative atomic and molecular masses.

Key Concepts

• Branches: physical, organic, inorganic, analytical, biochemistry
• Element, compound, mixture: definitions and examples
• Atom, molecule, ion, free radical
• Atomic number (Z) and mass number (A)
• Isotopes and their uses

Teaching Methodology (Step by Step)

1. Show samples: iron nail, water, air; ask students to classify them.
2. Build the element-compound-mixture idea with particle diagrams.
3. Explain isotopes using hydrogen (protium, deuterium, tritium).
4. Practise calculating molecular masses of common compounds.

Classroom Activities

• Classify ten household substances as element, compound or mixture.
• Draw particle diagrams for an element, a compound and a mixture.

Assessment Questions

1. Define an element and a compound with one example each.
2. What are isotopes? Name the isotopes of hydrogen.
3. Calculate the molecular mass of H2SO4.
4. Name any four branches of chemistry.

Homework

• Complete the classification table in the notebook.
• Calculate molecular masses of five given compounds.

Time Allocation

5 periods of 40 minutes each

Unit 2: Structure of Atoms

Learning Objectives

• Describe the discoveries of electron, proton and neutron.

• Explain Rutherford's and Bohr's atomic models.

• Write electronic configurations of the first 20 elements.

• Define shells, subshells and orbitals.

Key Concepts

• Cathode rays: discovery of the electron

• Proton and neutron: properties and location

• Rutherford's nuclear model and its defects

• Bohr's model: orbits and energy levels

• Electronic configuration and the Aufbau idea

Teaching Methodology (Step by Step)

1. Tell the story of the cathode-ray experiment as a discovery tale.
2. Compare Rutherford's and Bohr's models with simple orbit diagrams.
3. Practise electronic configurations on the board with student volunteers.
4. Use the first-20-elements chart for quick oral drills.

Classroom Activities

• Build atom models of the first ten elements with beads or clay.

• Quiz in pairs: one names an element, the other gives its configuration.

Assessment Questions

1. Who discovered the electron and how?
2. Write the electronic configuration of calcium (Z = 20).
3. State two defects of Rutherford's atomic model.
4. What is the maximum number of electrons in the M shell?

Homework

• Write configurations of elements 1 to 20 in the notebook.

• Draw a labelled diagram of Bohr's model for sodium.

Time Allocation

6 periods of 40 minutes each

Unit 3: Periodic Table and Periodicity

Learning Objectives

• State the modern periodic law.

• Describe the arrangement of periods and groups.

• Explain trends in atomic size, ionization energy and electronegativity.

• Classify elements as metals, non-metals and metalloids.

Key Concepts

• Modern periodic law: properties repeat with atomic number

• Periods (1-7) and groups (1-18)

• Metals, non-metals and metalloids on the table

• Shielding effect and effective nuclear charge

• Periodic trends across a period and down a group

Teaching Methodology (Step by Step)

1. Hand out a blank periodic table and let students locate familiar elements.
2. Explain the logic of periods and groups using electronic configuration.
3. Draw trend arrows on a large table chart for size and ionization energy.
4. Give a short oral quiz on group numbers of common elements.

Classroom Activities

• Colour-code a printed periodic table: metals, non-metals, metalloids.

• Predict the group and period of an element from its configuration.

Assessment Questions

1. State the modern periodic law.
2. How does atomic size change across a period? Why?
3. Name two metalloids and write their symbols.
4. In which group are the noble gases placed?

Homework

• Draw the first three periods and label one metal, one non-metal.

• Write short notes on any two periodic trends.

Time Allocation

5 periods of 40 minutes each

Unit 4: Structure of Molecules

Learning Objectives

• Explain how ionic and covalent bonds are formed.

• Draw dot-and-cross diagrams for simple molecules.

• Distinguish between single, double and triple covalent bonds.

• Explain coordinate covalent (dative) bonding with examples.

Key Concepts

• Octet and duplet rules

• Ionic bond: transfer of electrons (NaCl, MgO)

• Covalent bond: sharing of electrons (H2, Cl2, CH4)

• Multiple bonds: O2 (double), N2 (triple)

• Coordinate bond: NH4+ ion formation

Teaching Methodology (Step by Step)

1. Act out electron transfer vs sharing with student volunteers as atoms.
2. Draw dot-and-cross diagrams step by step for NaCl and H2.
3. Build molecule models with balls and sticks for CH4 and H2O.
4. Compare properties of ionic and covalent compounds in a table.

Classroom Activities

• Make dot-and-cross diagrams for five given molecules in pairs.

• Sort compound cards into ionic and covalent groups.

Assessment Questions

1. How is an ionic bond different from a covalent bond?
2. Draw the dot-and-cross diagram of magnesium oxide.
3. What is a coordinate covalent bond? Give one example.
4. Why do ionic compounds conduct electricity when molten?

Homework

• Draw dot-and-cross diagrams for HCl, O2 and NH3.

• Write five properties each of ionic and covalent compounds.

Time Allocation

6 periods of 40 minutes each

Unit 5: Physical States of Matter

Learning Objectives

• Describe the three states of matter using kinetic molecular theory.

• Explain diffusion, effusion and compressibility of gases.

• State Boyle's law and Charles's law with graphs.

• Explain vapour pressure and boiling point.

Key Concepts

• Kinetic molecular theory of gases

• Diffusion and effusion; Graham's law idea

• Boyle's law: P1V1 = P2V2

• Charles's law: V1/T1 = V2/T2

• Boiling point, vapour pressure and their factors

Teaching Methodology (Step by Step)

1. Spray perfume at one corner: ask how the smell reaches everyone.
2. Demonstrate gas compression with a syringe.
3. Plot P-V and V-T graphs from sample data on the board.
4. Explain why water boils faster on mountains.

Classroom Activities

• Time how long a scent takes to cross the room; discuss variables.

• Solve Boyle's and Charles's law numericals in small groups.

Assessment Questions

1. State Boyle's law with its mathematical form.
2. Why do gases diffuse faster than liquids?
3. Define boiling point.
4. A gas occupies 2 L at 1 atm. Find its volume at 2 atm (constant T).

Homework

• Solve five numericals on gas laws.

• Write the postulates of kinetic molecular theory.

Time Allocation

6 periods of 40 minutes each

Unit 6: Solutions

Learning Objectives

• Define solution, solute, solvent and concentration.

• Express concentration as percentage, molarity and mole fraction.

• Explain solubility and factors affecting it.

• Distinguish between saturated, unsaturated and supersaturated solutions.

Key Concepts

• True solution vs suspension vs colloid

• Concentration units: % w/w, % v/v, molarity

• Solubility and solubility curves

• Effect of temperature on solubility

• Like-dissolves-like rule

Teaching Methodology (Step by Step)

1. Dissolve sugar and sand in water: compare what happens.
2. Prepare solutions of known percentage concentration as a demo.
3. Show a solubility curve chart and read values from it.
4. Discuss why salt dissolves in water but oil does not.

Classroom Activities

• Prepare 100 mL of a 10% sugar solution and label it.

• Test solubility of salt in cold vs hot water and record results.

Assessment Questions

1. Define molarity with its formula.
2. What is a saturated solution?
3. Calculate the molarity of a solution with 2 moles in 4 L.
4. Name two factors that affect solubility.

Homework

• Solve numericals on percentage and molar concentrations.

• Draw a solubility curve and explain its use.

Time Allocation

5 periods of 40 minutes each

Unit 7: Electrochemistry

Learning Objectives

• Define oxidation and reduction in terms of electrons.

• Explain oxidizing and reducing agents with examples.

• Describe the construction of a simple electrochemical cell.

• Explain electrolysis of water with a labelled diagram.

Key Concepts

• Oxidation: loss of electrons; Reduction: gain of electrons

• OILRIG memory rule

• Oxidizing agents (gain electrons) vs reducing agents (lose electrons)

• Galvanic cell: anode, cathode, salt bridge

• Electrolysis of water: H2 and O2 in 2:1 ratio

Teaching Methodology (Step by Step)

1. Use OILRIG to fix oxidation-reduction ideas with quick examples.
2. Build a simple lemon/fruit cell and light a small LED if available.
3. Draw the electrolysis-of-water apparatus step by step.
4. Identify the oxidized and reduced species in given reactions.

Classroom Activities

• Set up a fruit battery in groups and measure its voltage.

• Classify ten given substances as oxidizing or reducing agents.

Assessment Questions

1. Define oxidation and reduction in electronic terms.
2. What gases are collected at each electrode in water electrolysis?
3. Name the anode and cathode in a galvanic cell.
4. Give one example each of an oxidizing and a reducing agent.

Homework

• Draw and label the electrolysis of water apparatus.

• Identify oxidation/reduction in five given chemical equations.

Time Allocation

5 periods of 40 minutes each

Unit 8: Chemical Reactivity

Learning Objectives

• Define metals and non-metals and compare their properties.

• Explain the reactivity series of metals.

• Describe reactions of metals with oxygen, water and acids.

• Explain corrosion and methods to prevent rusting.

Key Concepts

• Physical and chemical properties of metals vs non-metals

• Reactivity series: K, Na, Ca, Mg, Al, Zn, Fe, Sn, Pb, Cu, Ag, Au

• Metal + oxygen, metal + water, metal + dilute acid

• Displacement reactions

• Rusting: conditions and prevention (painting, oiling, galvanizing)

Teaching Methodology (Step by Step)

1. Show metal samples: ask students to list observable properties.
2. Demonstrate the reaction of sodium or magnesium with water carefully.
3. Use the reactivity series chart to predict displacement reactions.
4. Discuss rusted objects: why does rusting need both air and water?

Classroom Activities

• Test reactions of zinc, iron and copper with dilute acid (teacher demo).

• Design a poster: five ways to prevent rusting.

Assessment Questions

1. Write the reactivity series of any six metals in order.
2. What are the necessary conditions for rusting?
3. Name two methods to prevent rusting.
4. Which metal is more reactive: zinc or copper? How do you know?

Homework

• Write equations for reactions of metals with oxygen and acids.

• Prepare a chart of the reactivity series for the notebook.

Time Allocation

5 periods of 40 minutes each